EXPERIMENT– 2 Calculator

PROJECT NAME

Simple Calculator Using Arduino UNO, 4x4 Keypad and SSD1306 OLED Display

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OBJECTIVE / PROBLEM STATEMENT

This project implements a simple calculator using Arduino UNO, a 4x4 keypad, and an SSD1306 OLED display. The calculator performs basic arithmetic operations such as addition, subtraction, multiplication, and division. The result is displayed on the OLED screen.

COMPONENTS USED

Component	Quantity
Arduino UNO	1
4x4 Keypad	1
SSD1306 OLED Display	1
Jumper Wires	Required
Breadboard	1

PIN CONNECTIONS

4x4 KEYPAD CONNECTIONS

Keypad Pin	Arduino Pin
R1	D1
R2	D2
R3	D3
R4	D4
C1	D5
C2	D6
C3	D7
C4	D8

Symbols mapped operators

Symbol	Operation
A	+
B	-
C	*
D	/
#	AC

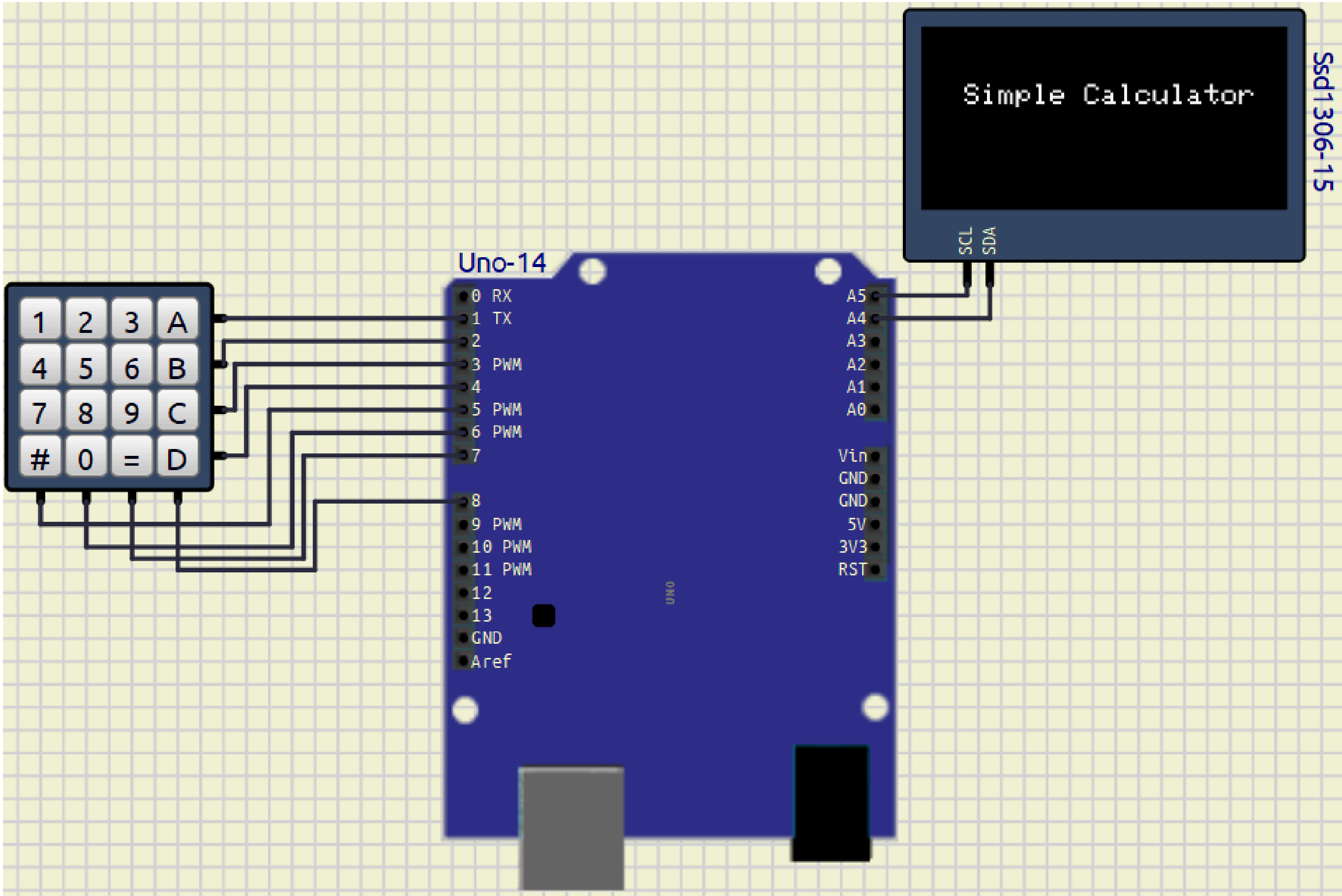
SSD1306 OLED DISPLAY CONNECTIONS

OLED Pin	Arduino Pin
VCC	5V
GND	GND
SDA	A4
SCL	A5

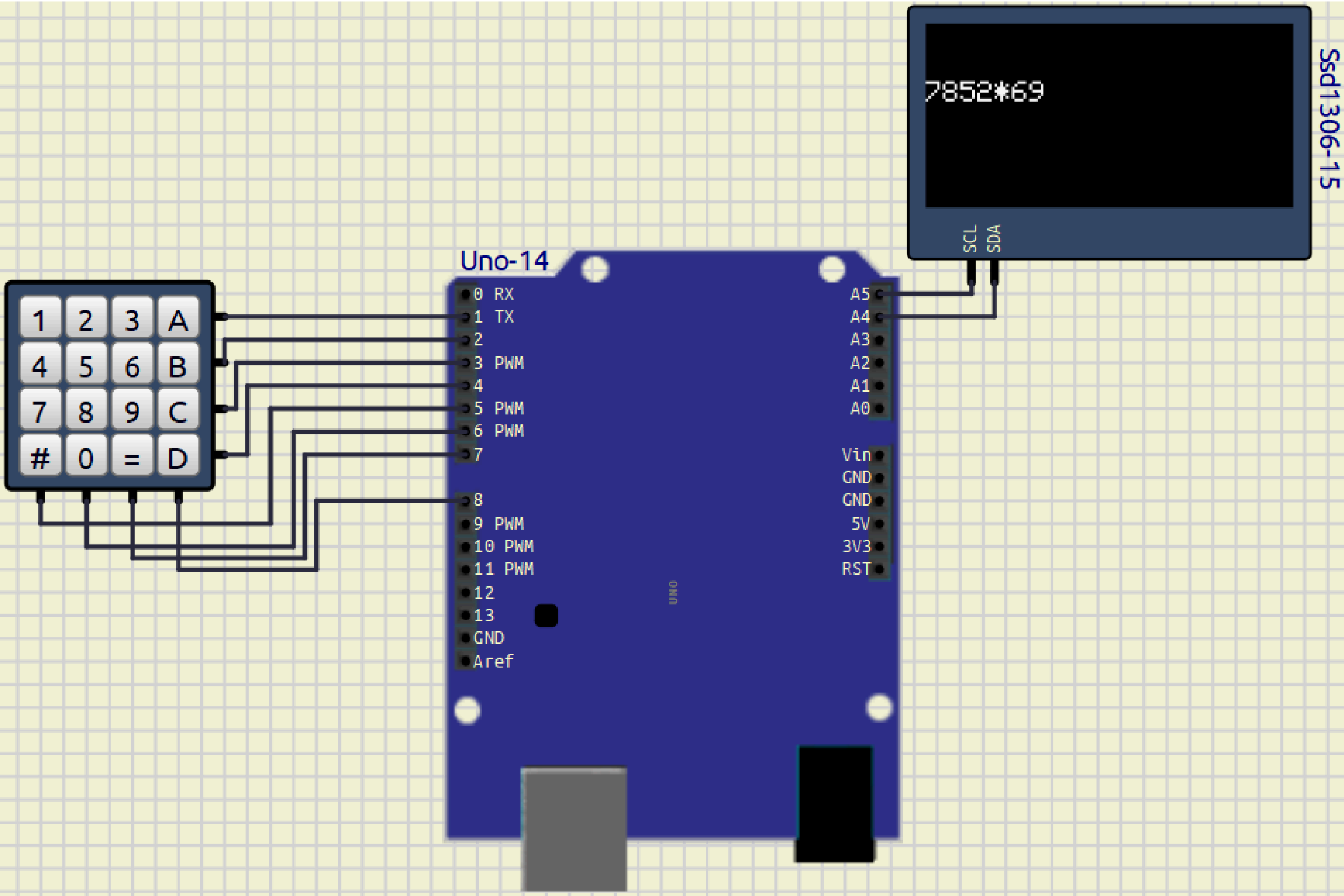
SOFTWARE USED

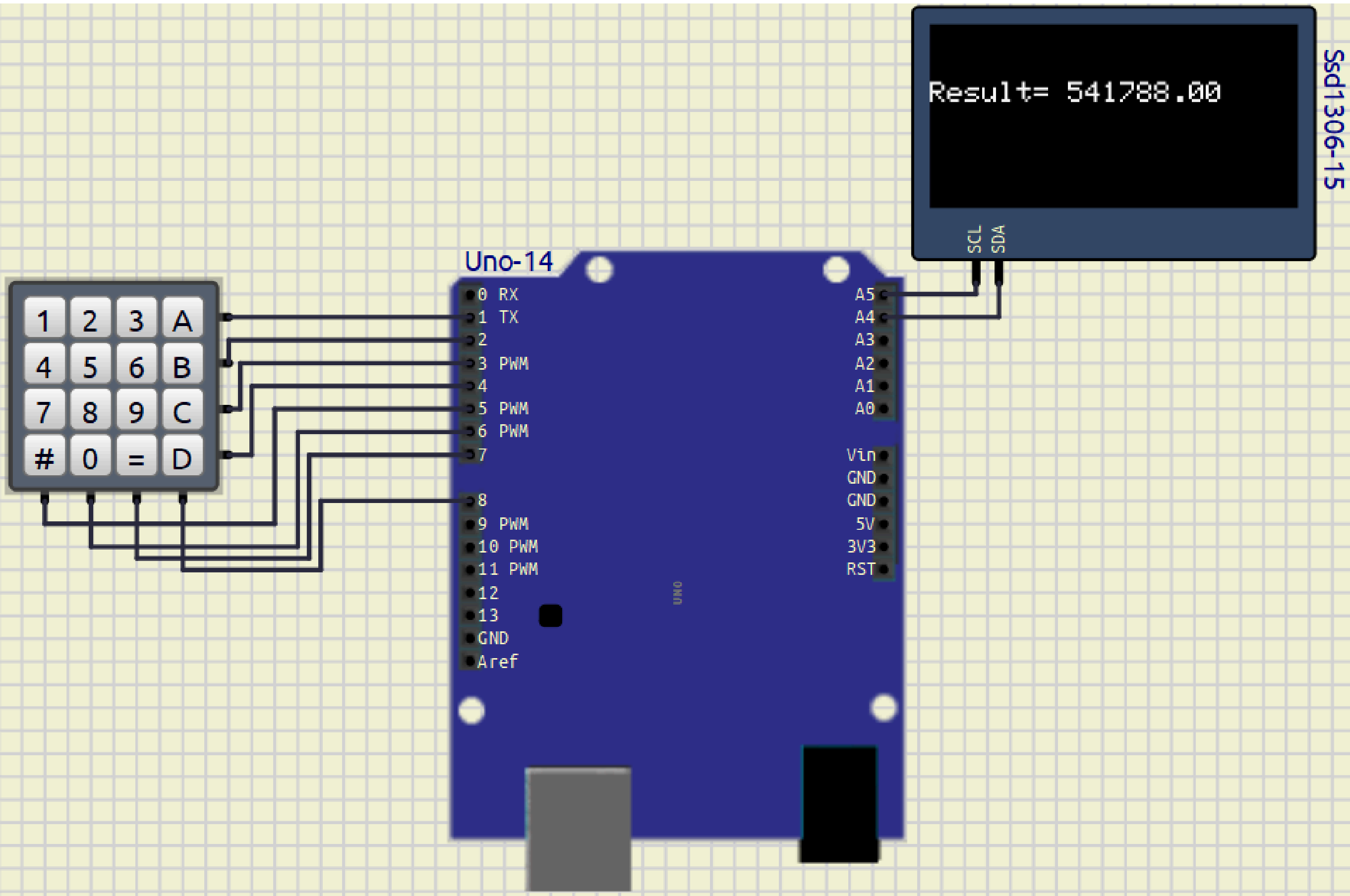
- Arduino IDE
- SimulIDE

CIRCUIT DIAGRAM


Simple Calculator
Ssd1306-15
SCL
SDA
Uno-14
0 RX
1 TX
2
3 PWM
4
5 PWM
6 PWM
7
8
9 PWM
10 PWM
11 PWM
12
13
GND
Aref
A5
A4
A3
A2
A1
A0
Vin
GND
GND
5V
3V3
RST
UNO
1 2 3 A
4 5 6 B
7 8 9 C
0 = D

Example:





REQUIRED LIBRARIES

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <Keypad.h>
```

MAIN ARDUINO CODE

Code With Comments ---> Click here

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <Keypad.h>

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

const byte ROWS = 4;
const byte COLS = 4;

char keys[ROWS][COLS] = {
  {'1', '2', '3', '+'},
  {'4', '5', '6', '-'},
  {'7', '8', '9', '*'},
  {'#', '0', '=', '/'}
};

byte rowPins[ROWS] = {1, 2, 3, 4};
byte colPins[COLS] = {5, 6, 7, 8};

Keypad keypad = Keypad(makeKeymap(keys),
                       rowPins,
                       colPins,
                       ROWS,
                       COLS);

String num1 = "";
String num2 = "";

char op;

float result;

bool operatorPressed = false;

String expression = "";
```

```
void setup()
{
    display.begin(SSD1306_SWITCHCAPVCC, 0x3C);

    display.clearDisplay();

    display.setTextSize(1);

    display.setCursor(0, 0);
    display.println("Simple Calculator");

    display.setCursor(0, 20);
    display.println("Can Perform");

    display.setCursor(0, 40);
    display.println("(+,-,*,/)");

    display.display();

    delay(2000);

    display.clearDisplay();
    display.display();
}

void loop()
{
    char key = keypad.getKey();

    if(key)
    {
        // CLEAR ALL
        if(key == '#')
        {
            num1 = "";
            num2 = "";
            expression = "";

            operatorPressed = false;

            result = 0;

            display.clearDisplay();
            display.display();
        }

        // NUMBER INPUT
        else if(isDigit(key))
        {
            if(!operatorPressed)
            {
                num1 += key;
            }
            else
```

```
            {
                num2 += key;
            }

            expression += key;

            updateDisplay(expression);
        }

        // OPERATOR INPUT
        else if(key == '+' || key == '-' ||
                key == '*' || key == '/')
        {
            if(!operatorPressed && num1 != "")
            {
                op = key;

                operatorPressed = true;

                expression += key;

                updateDisplay(expression);
            }
        }

        // RESULT
        else if(key == '=')
        {
            float n1 = num1.toFloat();
            float n2 = num2.toFloat();

            switch(op)
            {
                case '+':
                    result = n1 + n2;
                    break;

                case '-':
                    result = n1 - n2;
                    break;

                case '*':
                    result = n1 * n2;
                    break;

                case '/':

                    if(n2 != 0)
                    {
                        result = n1 / n2;
                    }
                    else
                    {
                        display.clearDisplay();
```

```
                        display.setTextSize(2);

                        display.setCursor(0, 20);

                        display.println("ERROR");

                        display.display();

                        delay(2000);

                        return;
                    }

                    break;
            }

            expression = "Result= " + String(result);

            updateDisplay(expression);

            num1 = String(result);

            num2 = "";

            operatorPressed = false;

            expression = String(result);
        }
    }
}

void updateDisplay(String text)
{
    display.clearDisplay();

    display.setTextSize(2);

    display.setCursor(0, 20);

    display.println(text);

    display.display();
}
```

WORKING PRINCIPLE

1. The 4x4 keypad is used to enter numbers and operators.
2. Arduino UNO continuously scans the keypad.
3. Entered values are stored as operands.
4. The selected operator determines the operation to be performed.

5. When '=' is pressed:
    - Arduino performs the calculation.
    - Result is displayed on the SSD1306 OLED display.
6. '#' key clears all previous inputs.
7. The process repeats continuously.

OUTPUT

Startup Screen

```
Simple Calculator
```

Example Calculation

```
5+3

Result= 8
```

FEATURES

- Performs Addition
- Performs Subtraction
- Performs Multiplication
- Performs Division
- OLED Display Output
- Clear Function using '#'
- Real-time Input Detection
- Divide by Zero Error Handling

ADVANTAGES

- Simple and easy to implement
- Low-cost embedded project
- Real-time calculations
- User-friendly interface
- Useful for learning keypad interfacing

APPLICATIONS

- Basic calculator systems
- Embedded system projects

- Learning Arduino interfacing
- Educational mini-projects
- Digital input-output systems

CONCLUSION

The Simple Calculator using Arduino UNO, 4x4 Keypad, and SSD1306 OLED Display was successfully implemented and tested. The system successfully performs basic arithmetic operations and displays the result on the OLED screen. During the experiment I faces problem in mapping those keys and assign them their values. I overcome it by the open source resources given in web and arduino forum

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REFERENCES

1. Arduino Keypad Library Documentation
   https://www.arduino.cc/reference/en/libraries/keypad/

2. SSD1306 OLED Display Library
   https://github.com/adafruit/Adafruit_SSD1306

3. Arduino UNO Official Documentation
   https://docs.arduino.cc/hardware/uno-rev3/

4. Adafruit GFX Graphics Library
   https://github.com/adafruit/Adafruit-GFX-Library

5. 4×4 Keypad Arduino Code, Pinout & Interfacing Complete Guide
   https://robosans.com/learn/embedded/arduino/4x4-keypad-arduino-code-pinout-interfacing-lcd/
